THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHEMATICS TWO-CHAPTER TEST SET A

Chapters 3 & 4: **Pair of Linear Equations in Two Variables, Quadratic Equations** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Use graph paper wherever a graph is asked.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The pair $3x + 4y = 5$ and $6x + 8y = 7$ has: 1

(a) two solutions **(b)** a unique solution
(c) no solution **(d)** infinitely many solutions

2. The solution of $2x + y = 7$ and $x - y = 2$ is: 1

(a) $(2, 3)$ **(b)** $(3, 1)$ **(c)** $(4, -1)$ **(d)** $(1, 3)$

3. The value of k for which $2x + 3y = 5$ and $4x + ky = 10$ have infinitely many solutions is: 1

(a) 6 **(b)** 5 **(c)** 3 **(d)** −6

4. A student states: "The lines $x = 2$ and $y = 2$ are parallel." The correct response is: 1

(a) Incorrect — they intersect at $(2, 2)$ **(b)** Correct
(c) They are coincident **(d)** Cannot be determined

5. To check whether $(1, 2)$ satisfies $3x + y = 5$ and $x + 2y = 5$, Student A says "both", Student B says "only the first". Then: 1

(a) Student A is correct **(b)** Student B is correct
(c) Both are correct **(d)** Both are wrong

6. The roots of $x^2 - 7x + 12 = 0$ are: 1

(a) $2, 6$ **(b)** $3, 4$ **(c)** $1, 12$ **(d)** $-3, -4$

7. The discriminant of $x^2 + 4x + 5 = 0$ is: 1

(a) 0 **(b)** 36 **(c)** −4 **(d)** 4

8. If $x^2 - 6x + k = 0$ has equal roots, then k is: 1

(a) 36 **(b)** 6 **(c)** 9 **(d)** 3

9. A student states: "$x^2 = 4x$ has only one root, 4." The correct response is: 1

(a) Its roots are ± 2 **(b)** It has no roots
(c) Incorrect — $x = 0$ is also a root **(d)** Correct

10. Which equation has two distinct real roots? 1

(a) $2x^2 - 3x + 4 = 0$ **(b)** $x^2 + x + 1 = 0$ **(c)** $x^2 - 5x + 6 = 0$ **(d)** $x^2 + 2x + 1 = 0$

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Solve by substitution: $2x + 3y = 11$ and $2x - 4y = -24$. 2

12. For which values of k will $kx - y = 2$ and $6x - 2y = 3$ have a unique solution? 2

13. Solve by factorisation: $3x^2 - 2\sqrt{6}x + 2 = 0$. 2

14. Find the nature of the roots of $2x^2 - 3x + 5 = 0$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. The sum of the numerator and denominator of a fraction is 12. If the denominator is increased by 3, the fraction becomes $\frac{1}{2}$. Find the fraction. 3

16. Solve $x - \frac{1}{x} = 3$ ($x \neq 0$) using the quadratic formula. 3

17. Solve graphically: $2x - y = 4$ and $x + y = -1$. 3

OR

In $\triangle ABC$, $\angle C = 3\angle B = 2(\angle A + \angle B)$. Find the three angles. 3

18. A natural number, when increased by 12, equals 160 times its reciprocal. Find the number. 3

OR

The sum of a number and its reciprocal is $\frac{50}{7}$. Find the number. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study:** A school canteen sells 2 samosas and 3 kachoris for ₹65, and 3 samosas and 2 kachoris for ₹60. 5

(i) Form the equations.

(ii) Find the price of each item.

(iii) Find the cost of 5 samosas and 4 kachoris.

OR

2 kg of apples and 1 kg of grapes cost ₹160 on a day. After a month, 4 kg of apples and 2 kg of grapes cost ₹300. Represent the situation algebraically and graphically, and state what you conclude. 5

20. **Case study:** A rectangular swimming pool is 20 m long and 10 m wide. A path of uniform width is built around it; the area of the path is 216 m². 5

(i) Form a quadratic equation.

(ii) Find the width of the path.

(iii) Find the cost of tiling the path at ₹50 per m².

OR

A plane left 30 minutes late. To reach its destination 1500 km away on time, it had to increase its usual speed by 100 km/h. Find its usual speed and verify. 5

END OF QUESTION PAPER — Total: 40 Marks

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